MATB24H3
Exercise Questions

Student Name: Your Name **Student Number: Your Student Number**

2.1.1. Consider the set $X = \mathbb{Q} \setminus \{0\}$ and the map $\star$ is depicted by $a \star b = \frac{a}{b}$. Is $\star$ a binary operation?

Solution:

2.1.2. Is $(\mathcal{P}(\mathbb{N}), \cap)$ a monoid? what is the identity element? How about $(\mathcal{P}(\mathbb{N}), \cup)$ for any arbitrary set X?

Solution:

2.2.1. Suppose d is square-free, so there is no integer m such that $m^2 = d$. Show that $(\mathbb{Q}(\sqrt{d}),\ +,\ \cdot)$ forms a field. What happens if $d = m^2$ for some m?

Solution:

2.2.2. Now, can we extend a finite field in the same way we extended an infinite field? If so, give an example.

Solution:

2.2.3. Prove that $0_{\mathbb{F}}$, $1_{\mathbb{F}}$ and the additive inverse element $-a$ of any element $a \in F$ are unique.

Solution:

2.2.4. Prove $(-a)(-b) = ab$ for all $a, b \in \mathbb{F}$.

Solution:

2.2.5. Prove that the triple $(\mathbb{Z}_p, + \mod p, \cdot \mod p)$ is a field $\iff$ p is a prime. Use the fact that if two integers m, n are relatively prime, then there exist integers r and s such that $mr + ns = 1$.

Solution:

2.2.6. Prove that the 2-tuple $(S, \circ)$, where S is the set of symmetries of a square and $\circ$ is the composition operation, does indeed form a group. Moreover, find the symmetry group of a rectangle.

Solution:

2.2.7. Characterise all idempotent elements of $(\mathcal{M}_{2\times 2}(\mathbb{R}), +, \cdot)$.

Solution:

2.2.8. Prove that $\mathcal{C}(\mathcal{R})$ is a subring of any ring $(\mathcal{R}, +, \cdot)$.

Solution:

2.2.9. Let $(\mathcal{R}, +, \cdot)$ be a ring and suppose that $\alpha \in \mathcal{R}$ has a multiplicative inverse. Moreover, suppose that $\alpha \in C(\mathcal{R})$. Is $\alpha^{-1} \in C(\mathcal{R})$?

Solution:

2.2.10. Does the set of functions with infinitely many roots form a subring of $\mathcal{F}(\mathbb{R})$? Justify your answer.

Solution:

2.2.11. [REFER TO NOTES]

Solution:

2.2.12. Determine the zero and the identity elements of the polynomial ring $(\mathcal{R}[x], +, \cdot)$ where $\mathcal{R} = \mathcal{C}(\mathcal{M}_{2\times 2}(\mathbb{R}))$.

Solution:

2.2.13. True/False. Justify:

1. **Let $\mathbb{F}$ be any field. Suppose that α and β are two elements of an extension field $\mathbb{K}$ of $\mathbb{F}$. Then $\mathbb{F}(\alpha)$ and $\mathbb{F}(\beta)$ are always distinct extension fields of $\mathbb{F}$.**
2. **Given any set X, there is no operation $\cdot$ that can be defined on $[X]^{\leq \|X\|}$ that makes it a group.**
3. **Given a monoid $(\mathcal{S}, *)$ there exists a binary operation $\cdot$ on $\mathcal{S}$ such that $(\mathcal{S}, \cdot)$ no longer admits an identity element. If so, what assumption did you need to make?**
4. **$(\mathbb{R}, \cdot)$ forms a group.**
5. **$(\mathbb{R}, +)$ forms a group with the identity element being 1.**

6. **Suppose $(\mathbb{F}, +, \cdot)$ is the field. Then $(\mathbb{F}, +)$ and $(\mathbb{F}, \cdot)$ both form a group.**

Solution:

3.1.1. Had we defined vector addition by $\hat{A} \boxplus \hat{B} = [a_{ij} \cdot b_{ij}]$, i.e., pointwise multiplication of the entries in Example 3.1.3, would $(\mathcal{M}_{m\times n}, \boxplus, \odot, (\mathbb{F}, +, \cdot))$ form a vector space over $\mathbb{F}$?

Solution:

3.1.2. Prove properties 2, 4 and 5 of Theorem 3.1.2, namely, for any vector space $V = (V, \boxplus, \odot, (\mathbb{F}, +, \cdot))$ over field $\mathbb{F}$:

2. **The additive inverse of any $v \in V$ is unique.**
4. **Any scalar of the zero vector yields the zero vector, i.e. $\forall\, \alpha \in \mathbb{F}$, $a \odot \vec{0} = \vec{0}$.**
5. **For all $v \in V$ and $c \in \mathbb{F}$, $(-c) \odot v = c \odot (-v) = -1 \odot (c \odot v)$.**

Solution:

3.2.1. Prove Theorem 3.2.2, namely if $S \subseteq V$ then $\text{span}(S) \subseteq V$ is a subspace of V.

Solution:

3.2.2. Prove Theorem 3.2.3, namely: Let J be any index set (It is a set whose elements are used to enumerate the elements of another set.) and let $\{W_\alpha \mid \alpha \in J\}$ be a collection of subspaces of V. Then $\bigcap_{\alpha \in J} W_\alpha$ is a subspace of V.

Solution:

3.2.3. Prove Theorem 3.2.4, namely: Let $W_1 \subseteq W_2 \subseteq W_3 \subseteq \dots$ be an increasing chain of subspaces of V. Then $\bigcup_{n\in\mathbb{N}} W_n$ is a subspace.

Solution:

3.3.1. Show that the set $\{e^x, e^{2x}, e^{3x}\}$ is linearly independent in the vector space of $\mathcal{F}(\mathbb{R})$ over $\mathbb{R}$.

Solution:

3.4.1. Prove Proposition 3.4.1, namely, the following:
Let $T\colon V \to W$ be a linear transformation from V to W. Then:

1. **For all $x, y \in V$, $T(x - y) = T(x) - T(y)$**
2. **T maps the zero vector $\vec{0}_V$ of V to the zero vector $\vec{0}_W$ of W.**
3. **If $T'\colon W \to Z$ is also a linear transformation, their composition $T' \circ T$ is also a linear transformation from V to Z where $T' \circ T = T'(T(v))$.**
4. **If V_1 is a subspace of V and W_1 is a subspace of W, then $T[V_1]$ is a subspace of W and $T^{-1}[W_1]$ is a subspace of V.**

Solution:

3.4.2. Prove Theorem 2.4.2 (currently under fix. please ignore this question)

Solution:

3.4.3. Prove Theorem 3.4.2, namely,
Show that $(\mathcal{L}(V, W), \boxplus, \odot, (\mathbb{F}, +\cdot))$ forms a vector space over $\mathbb{F}$, where $\boxplus$ and $\odot$ are the pointwise addition of linear maps and scalar multiplication with the linear map in W, respectively.
That is, $\mathcal{L}(V, W)$ is closed under both $\boxplus$ and $\odot$. That is to show that $T \boxplus S$ and $\alpha \odot T$ are indeed linear transformations from V to W and all the axioms of a vector space are satisfied.

Solution:

3.4.4. Prove that a linear map $T\colon V \to W$ is invertible $\iff$ T is injective and surjective.

Solution:

3.4.5. Let V and W be two vector spaces over a field $\mathbb{F}$ with dimensions n and $m > 3$ respectively. Construct three different linearly independent maps from V to W.

Solution:

3.4.6. Prove whether or not the triple $(\mathcal{L}(\mathbb{R}, \mathbb{R}), +, \cdot)$ forms a subring of $\mathcal{F}(\mathbb{R})$ under point-wise addition and multiplication.

Solution: